

Q1.

A particle A , of mass 0.2 kg , collides with a particle B , of mass 0.3 kg

Immediately before the collision, the velocity of A is $\begin{bmatrix} 4 \\ 12 \end{bmatrix} \text{ ms}^{-1}$

and the velocity of B is $\begin{bmatrix} -1 \\ -3 \end{bmatrix} \text{ ms}^{-1}$

As a result of the collision the particles coalesce to become a single particle.

Find the velocity of the single particle.

Circle your answer.

$\begin{bmatrix} 0.5 \\ 1.5 \end{bmatrix} \text{ ms}^{-1}$
 $\begin{bmatrix} 2 \\ 6 \end{bmatrix} \text{ ms}^{-1}$
 $\begin{bmatrix} 1 \\ 3 \end{bmatrix} \text{ ms}^{-1}$
 $\begin{bmatrix} 3 \\ 9 \end{bmatrix} \text{ ms}^{-1}$

(Total 1 mark)

Q2.

Two smooth spheres A and B of equal radius are free to move on a smooth horizontal surface.

The masses of A and B are m and $4m$ respectively.

The coefficient of restitution between the spheres is e .

The spheres are projected directly towards each other, each with speed u , and subsequently collide.

(a) Show that the speed of B immediately after the impact with A is

$$\frac{u(3-2e)}{5}$$

(4)

(b) Find the speed of A in terms of u and e .

(2)

(c) Comment on the direction of motion of the spheres after the collision, justifying your answer.

(2)

(d) The magnitude of the impulse on B due to the collision is I .

Deduce that

$$\frac{8mu}{5} \leq I \leq \frac{16mu}{5}$$

(3)

Q3.

Two discs, A and B , have equal radii and masses 0.8 kg and 0.4 kg respectively. The discs are placed on a horizontal surface.

The discs are set into motion when they are 3 metres apart, so that they move directly towards each other, each travelling at a speed of 6 m s^{-1} . The discs collide directly with each other.

After the collision A moves in the opposite direction with a speed of 1.2 m s^{-1}

The coefficient of restitution between the two discs is e .

- (a) Assuming that the surface is smooth, show that $e = 0.8$.

(5)

- (b) Describe one way in which the model you have used could be refined.

(1)

(Total 6 marks)

Q4.

Two particles, A and B , are moving towards each other along the same straight horizontal line when they collide. Particle A has mass 5 kg and particle B has mass 4 kg . Just before the collision, the speed of A is 4 m s^{-1} and the speed of B is 3 m s^{-1} . After the collision, the speed of A is 0.6 m s^{-1} and both particles move on the same straight horizontal line.

Find the two possible speeds of B after the collision.

(Total 6 marks)

Q5.

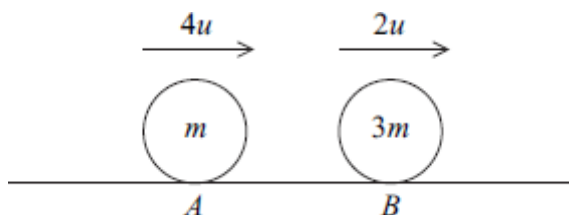
A toy train of mass 300 grams is moving along a straight horizontal track at a speed of 2.8 m s^{-1} . This toy train collides with another toy train, of mass 200 grams , which is at rest on the same track. During the collision, the two trains lock together and then move together.

Find the speed of the trains immediately after the collision.

(Total 3 marks)

Q6.

A smooth sphere A , of mass m , is moving with speed $4u$ in a straight line on a smooth horizontal table. A smooth sphere B , of mass $3m$, has the same radius as A and is moving on the table with speed $2u$ in the same direction as A .



The sphere A collides directly with sphere B . The coefficient of restitution between A and B is e .

(a) Find, in terms of u and e , the speeds of A and B immediately after the collision. (6)

(b) Show that the speed of B after the collision cannot be greater than $3u$. (2)

(c) Given that $e = \frac{2}{3}$, find, in terms of m and u , the magnitude of the impulse exerted on B in the collision. (3)

(Total 11 marks)

Q7.

Two particles, A of mass 7 kg and B of mass 3 kg, are moving on a smooth horizontal plane when they collide. Just before the collision, the velocity of A is $(3\mathbf{i} + 8\mathbf{j}) \text{ m s}^{-1}$ and the velocity of B is $(6\mathbf{i} - 5\mathbf{j}) \text{ m s}^{-1}$. During the collision, the particles coalesce to form a single combined particle.

Find the velocity of the single combined particle after the collision.

(Total 3 marks)

Q8.

Two toy trains, A and B , are moving in the same direction on a straight horizontal track when they collide. As they collide, the speed of A is 4 m s^{-1} and the speed of B is 3 m s^{-1} . Immediately after the collision, they move together with a speed of 3.8 m s^{-1} .

The mass of A is 2 kg. Find the mass of B .

(Total 3 marks)

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Q9.

Two toy trains, A and B , are moving in the same direction on a straight horizontal track when they collide. As they collide, the speed of A is 4 m s^{-1} and the speed of B is 3 m s^{-1} . Immediately after the collision, they move together with a speed of 3.8 m s^{-1} .

The mass of A is 2 kg. Find the mass of B .

(Total 3 marks)

Q10.

A trolley, of mass 5 kg, is moving in a straight line on a smooth horizontal surface. It has a velocity of 6 m s^{-1} when it collides with a stationary trolley, of mass m kg. Immediately after the collision, the trolleys move together with velocity 2.4 m s^{-1} .

Find m .

(Total 3 marks)

Q11.

Two particles, A and B , are moving on a smooth horizontal surface when they collide. The mass of A is 6 kg and the mass of B is $m \text{ kg}$. Before the collision, the velocity of A is $(5\mathbf{i} + 18\mathbf{j}) \text{ m s}^{-1}$ and the velocity of B is $(2\mathbf{i} - 5\mathbf{j}) \text{ m s}^{-1}$. After the collision, the velocity of A is $8\mathbf{i} \text{ m s}^{-1}$ and the velocity of B is $V\mathbf{j} \text{ m s}^{-1}$.

(a) Find m .

(3)

(b) Find V .

(3)

(Total 6 marks)

Q12.

Two particles, A and B , are travelling in the same direction along a straight line on a smooth horizontal surface. Particle A has mass 3 kg and particle B has mass 7 kg . Particle A has a speed of 20 m s^{-1} and particle B has a speed of 10 m s^{-1} , as shown in the diagram.



Particle A and particle B collide and coalesce to form a single particle. Find the speed of this single particle after the collision.

(Total 3 marks)

Q13.

Two particles, A and B , are moving on a smooth horizontal plane when they collide.

The mass of A is 6 kg and the mass of B is $m \text{ kg}$. Before the collision, the velocity of A

is $\begin{bmatrix} 2 \\ 4 \end{bmatrix} \text{ m s}^{-1}$ and the velocity of B is $\begin{bmatrix} 3 \\ -2 \end{bmatrix} \text{ m s}^{-1}$. After the collision, the velocity of A is $\begin{bmatrix} 1 \\ 3 \end{bmatrix} \text{ m s}^{-1}$ and the velocity of B is $\begin{bmatrix} 7 \\ b \end{bmatrix} \text{ m s}^{-1}$.

(a) Find m .

(4)

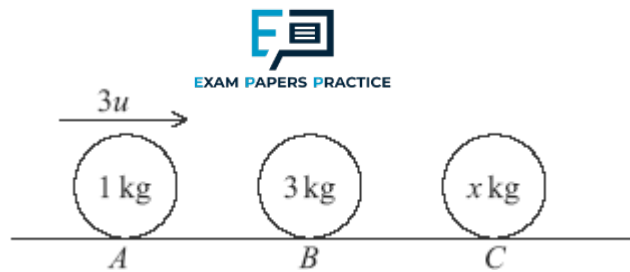
(b) Find b .

(2)

(Total 6 marks)

Q14.

Three smooth spheres, A , B and C , of equal radii have masses 1 kg , 3 kg and $x \text{ kg}$ respectively. The spheres lie at rest in a straight line on a smooth horizontal surface with B between A and C . The sphere A is projected with speed $3u$ directly towards B and collides with it.



The coefficient of restitution between each pair of spheres is $\frac{1}{3}$.

- (a) Show that A is brought to rest by the impact and find the speed of B immediately after the collision in terms of u .

(6)

- (b) Subsequently, B collides with C .

Show that the speed of C immediately after the collision is $\frac{4u}{3+x}$.

Find the speed of B immediately after the collision in terms of u and x .

(6)

- (c) Show that B will collide with A again if $x > 9$.

(2)

- (d) Given that $x = 5$, find the magnitude of the impulse exerted on C by B in terms of u .

(2)

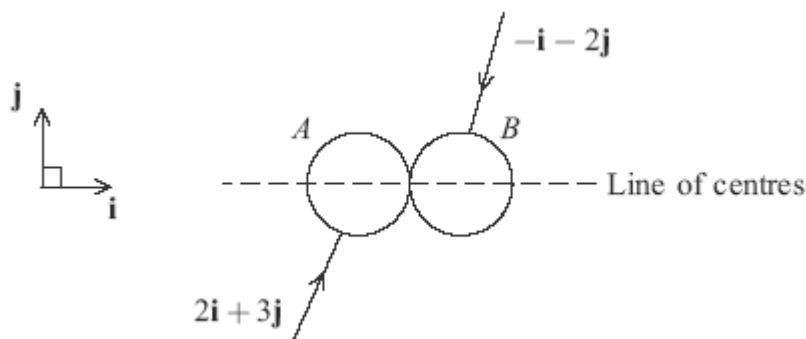
(Total 16 marks)

Q15.

Two smooth spheres, A and B , have equal radii and masses 1 kg and 2 kg respectively.

The sphere A is moving with velocity $(2\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$ and the sphere B is moving with velocity $(-\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$ on the same smooth horizontal surface.

The spheres collide when their line of centres is parallel to the unit vector $\mathbf{i}$, as shown in the diagram.



- (a) Briefly state why the components of the velocities of A and B parallel to the unit vector $\mathbf{j}$ are not changed by the collision.

(1)

- (b) The coefficient of restitution between the spheres is 0.5.

Find the velocities of A and B immediately after the collision.

(6)

(Total 7 marks)

Q16.

A particle of mass 4 kg is travelling at 8 m s^{-1} along a straight line when it collides with a stationary particle of mass 1 kg. After the collision, the two particles move together at the same speed.

Find the speed of the particles after the collision.

(Total 3 marks)

Q17.

Two particles, A and B , are travelling in the same direction with constant speeds along a straight line when they collide. Particle A has mass 2.5 kg and speed 12 m s^{-1} . Particle B has mass 1.5 kg and speed 4 m s^{-1} . After the collision, the two particles move together at the same speed.

Find the speed of the particles after the collision.

(Total 3 marks)

Q18.

Two particles, A and B , are moving on a smooth horizontal surface when they collide. During the collision, the two particles coalesce to form a single combined particle. Particle A has mass 3 kg and particle B has mass 7 kg.

Before the collision, the velocity of A is $\begin{bmatrix} 6 \\ -2 \end{bmatrix} \text{ m s}^{-1}$ and the velocity of B is $\begin{bmatrix} -1 \\ 4 \end{bmatrix} \text{ m s}^{-1}$.

- (a) Find the velocity of the combined particle after the collision.

(3)

- (b) Find the speed of the combined particle after the collision.

(2)

(Total 5 marks)

Q19.

A smooth sphere A of mass m is moving with speed $5u$ in a straight line on a smooth horizontal table. The sphere A collides directly with a smooth sphere B of mass $7m$, having the same radius as A and moving with speed u in the same direction as A . The coefficient of restitution between A and B is e .



- (a) Show that the speed of B after the collision is $\frac{u}{2}(e+3)$. (5)
- (b) Given that the direction of motion of A is reversed by the collision, show that $e > \frac{3}{7}$. (4)
- (c) Subsequently, B hits a wall fixed at right angles to the direction of motion of A and B . The coefficient of restitution between B and the wall is $\frac{1}{2}$. Given that after B rebounds from the wall both spheres move in the same direction and collide again, show also that $e < \frac{9}{13}$. (4)
- (Total 13 marks)

Q20.

Two particles, A and B , are moving on a horizontal plane when they collide and coalesce to form a single particle. The mass of A is 5 kg and the mass of B is 15 kg.

Before the collision, the velocity of A is $\begin{bmatrix} 2U \\ U \end{bmatrix}$ m s⁻¹ and the velocity of B is $\begin{bmatrix} V \\ -1 \end{bmatrix}$ m s⁻¹.

After the collision, the velocity of the combined particle is $\begin{bmatrix} V \\ 0 \end{bmatrix}$ m s⁻¹.

Find:

- (a) U ; (3)
- (b) V . (2)

(Total 5 marks)

Q21.

Two particles, A and B , are moving on a horizontal plane when they collide and coalesce to form a single particle. The mass of A is 5 kg and the mass of B is 15 kg.

Before the collision, the velocity of A is $\begin{bmatrix} 2U \\ U \end{bmatrix}$ m s⁻¹ and the velocity of B is $\begin{bmatrix} V \\ -1 \end{bmatrix}$ m s⁻¹.

After the collision, the velocity of the combined particle is $\begin{bmatrix} V \\ 0 \end{bmatrix}$ m s⁻¹.

(a) Find:

- (i) U ;

(3)

(ii) V .

(2)

(b) Find the speed of A before the collision.

(2)

(Total 7 marks)

Q22.

Two particles, A and B , are travelling towards each other along a straight horizontal line.

Particle A has velocity 2 m s^{-1} and mass $m \text{ kg}$.

Particle B has velocity -2 m s^{-1} and mass 3 kg .



The particles collide.

(a) If the particles move in opposite directions after the collision, each with speed 0.5 m s^{-1} , find the value of m .

(3)

(b) If the particles coalesce during the collision, forming a single particle which moves with speed 0.5 m s^{-1} , find the two possible values of m .

(5)

(Total 8 marks)

Q23.

Two smooth spheres, A and B , have equal radii and masses m and $2m$ respectively. The spheres are moving on a smooth horizontal plane. The sphere A has velocity $(4\mathbf{i} + 3\mathbf{j})$ when it collides with the sphere B which has velocity $(-2\mathbf{i} + 2\mathbf{j})$. After the collision, the velocity of B is $(\mathbf{i} + \mathbf{j})$.

(a) Find the velocity of A immediately after the collision.

(3)

(b) Find the angle between the velocities of A and B immediately after the collision.

(3)

(c) Find the impulse exerted by B on A .

(3)

(d) State, as a vector, the direction of the line of centres of A and B when they collide.

(1)

(Total 10 marks)

Q24.

Two particles A and B have masses of 3 kg and 2 kg respectively. They are moving along a straight horizontal line towards each other. Each particle is moving with a speed of 4 m s^{-1} when they collide.



- (a) If the particles coalesce during the collision to form a single particle, find the speed of the combined particle after the collision.

(3)

- (b) If, after the collision, A moves in the same direction as before the collision with speed 0.4 m s^{-1} , find the speed of B after the collision.

(3)

(Total 6 marks)

Q25.

Two particles, A and B , are moving on a smooth horizontal surface. Particle A has

mass 2 kg and velocity $\begin{bmatrix} 3 \\ -2 \end{bmatrix} \text{ m s}^{-1}$. Particle B has mass 3 kg and velocity $\begin{bmatrix} -4 \\ 1 \end{bmatrix} \text{ m s}^{-1}$.

The two particles collide, and they coalesce during the collision.

- (a) Find the velocity of the combined particles after the collision.

(3)

- (b) Find the speed of the combined particles after the collision.

(2)

(Total 5 marks)

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Q26.

Two small smooth spheres, A and B , of equal radii have masses 0.3 kg and 0.2 kg respectively. They are moving on a smooth horizontal surface directly towards each other with speeds 3 m s^{-1} and 2 m s^{-1} respectively when they collide. The coefficient of restitution between A and B is 0.8.

- (a) Find the speeds of A and B immediately after the collision.

(6)

- (b) Subsequently, B collides with a fixed smooth vertical wall which is at right angles to the path of the sphere. The coefficient of restitution between B and the wall is 0.7.

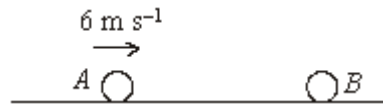
Show that B will collide again with A .

(3)

(Total 9 marks)

Q27.

A particle A moves across a smooth horizontal surface in a straight line. The particle A has mass 2 kg and speed 6 m s^{-1} . A particle B , which has mass 3 kg , is at rest on the surface. The particle A collides with the particle B .



- (a) If, after the collision, A is at rest and B moves away from A , find the speed of B . (3)
- (b) If, after the collision, A and B move away from each other with speeds $v \text{ m s}^{-1}$ and $4v \text{ m s}^{-1}$ respectively, as shown in the diagram below, find the value of v .



(3)
(Total 6 marks)

Q28.

Two particles, A and B , are moving on a smooth horizontal surface.

The particle A has mass $m \text{ kg}$ and is moving with velocity $\begin{bmatrix} 5 \\ -3 \end{bmatrix} \text{ m s}^{-1}$.

The particle B has mass 0.2 kg and is moving with velocity $\begin{bmatrix} 2 \\ 3 \end{bmatrix} \text{ m s}^{-1}$.

- (a) Find, in terms of m , an expression for the total momentum of the particles. (2)

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- (b) The particles A and B collide and form a single particle C , which moves with velocity $\begin{bmatrix} k \\ 1 \end{bmatrix} \text{ m s}^{-1}$, where k is a constant.

- (i) Show that $m = 0.1$. (3)

- (ii) Find the value of k . (3)

(3)
(Total 8 marks)

Q29.

Three smooth spheres A , B and C of equal radii and masses m , m and $2m$ respectively lie at rest on a smooth horizontal table. The centres of the spheres lie in a straight line with B between A and C . The coefficient of restitution between any two spheres is e .

The sphere A is projected directly towards B with speed u and collides with B .

- (a) Find, in terms of u and e , the speed of B immediately after the impact between A and B .

(5)

- (b) The sphere B subsequently collides with C . The speed of C immediately after this collision is $\frac{3}{8}u$. Find the value of e .

(7)

(Total 12 marks)

Q30.

A particle, A , of mass 12 kg is moving on a smooth horizontal surface with velocity $\begin{bmatrix} 4 \\ 7 \end{bmatrix}$ m s⁻¹.

It then collides and coalesces with a second particle, B , of mass 4 kg.

- (a) If before the collision the velocity of B was $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ m s⁻¹, find the velocity of the combined particle after the collision.

(4)

- (b) If after the collision the velocity of the combined particle is $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$ m s⁻¹, find the velocity of B before the collision.

(3)

(Total 7 marks)

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Q31.

Two particles, A and B , are moving in the same direction along a straight line. Particle A has mass 8 kg and speed 8 m s⁻¹. Particle B has mass 12 kg and speed 6 m s⁻¹. The two particles collide.



- (a) If, during the collision, the two particles coalesce, find the speed of the combined particle.

(3)

- (b) If, after the collision, particle A continues to move in the same direction at 6.5 m s⁻¹, find the speed of B after the collision.

(3)

(Total 6 marks)

Q32.

A particle of mass m has velocity $\begin{bmatrix} 4 \\ 2 \end{bmatrix}$ m s⁻¹. It then collides with a particle of mass 3 kg which has velocity $\begin{bmatrix} -1 \\ -1 \end{bmatrix}$ m s⁻¹. During the collision the particles coalesce and move with velocity $\begin{bmatrix} 1 \\ V \end{bmatrix}$ m s⁻¹.

- (a) Show that $m = 2$.

(4)

- (b) Find V .

(3)

(Total 7 marks)

Q33.

Three smooth spheres, A , B and C , of equal radii lie at rest in a straight line on a smooth horizontal table. Sphere A has mass $4m$ and is set in motion with velocity u . It collides directly with sphere B which has mass $2m$. The coefficient of restitution between A and B is e .

- (a) Find, in terms of e and u , the velocities of A and B immediately after the collision.

(5)

- (b) Given that the magnitude of the impulse exerted by A on B is $\frac{12mu}{5}$, show that

$$e = \frac{4}{5}$$

(3)

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- (c) After the collision between A and B , the sphere B collides directly with the sphere C which has mass m . The coefficient of restitution between B and C is $\frac{4}{5}$.

Find, in terms of u , the velocity of B after this collision.

(4)

- (d) Determine whether or not A and B will collide again. Give a reason for your answer.

(2)

(Total 14 marks)

Q34.

A particle A , of mass 0.1 kg, collides with a particle B , of mass 0.2 kg.

Immediately before the collision, the velocity of A is $\begin{bmatrix} 8 \\ 12 \end{bmatrix}$ m s⁻¹ and the velocity of B

is $V \text{ m s}^{-1}$.

Immediately after the collision, the velocity of A is $\begin{bmatrix} -2 \\ 6 \end{bmatrix} \text{ m s}^{-1}$ and the velocity of B is $\begin{bmatrix} 6 \\ 0 \end{bmatrix} \text{ m s}^{-1}$.

(a) Find V .

(4)

(b) The collision occurs at the origin, O . After the collision, the particles continue to move with constant velocities.

(i) Find the position vectors of A and B two seconds after the collision occurs.

(2)

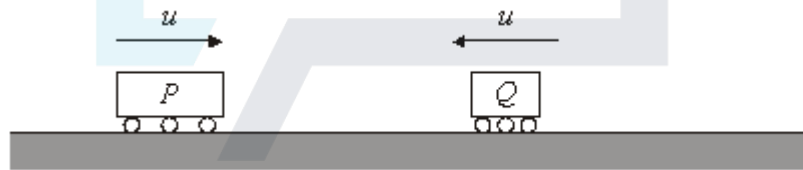
(ii) Find the distance between A and B two seconds after the collision occurs.

(3)

(Total 9 marks)

Q35.

Two toy train carriages, P and Q , have masses $2m$ and m respectively. They are moving directly towards each other, as shown in the diagram. Both carriages have speed u .



The carriages collide and subsequently Q moves in the same direction as P . The coefficient of restitution between P and Q is e .

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(a) (i) Show that the speed of carriage P immediately after the collision is $\frac{u}{3} (1 - 2e)$ and find the speed of carriage Q .

(7)

(ii) Deduce that $e < \frac{1}{2}$.

(2)

(b) The magnitude of the impulse on P due to the collision is I . Show that:

(i) $I = \frac{4mu}{3} (1 + e)$;

(3)

(ii) $\frac{4mu}{3} \leq I < 2mu$.

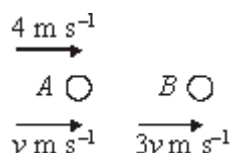
(2)

Q36.

A particle A of mass $3m$ kg is moving with speed $4m$ s in a straight line on a smooth horizontal surface.

The particle collides with a stationary particle B of mass m kg.

Immediately after the collision, A is moving with speed v m s⁻¹, and B is moving with speed $3v$ m s⁻¹ in the same direction.

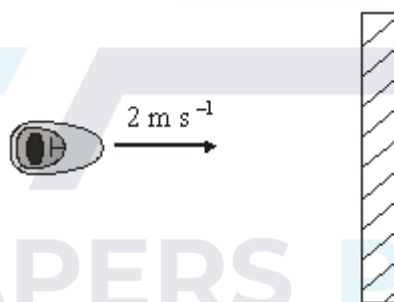


Find the value of v .

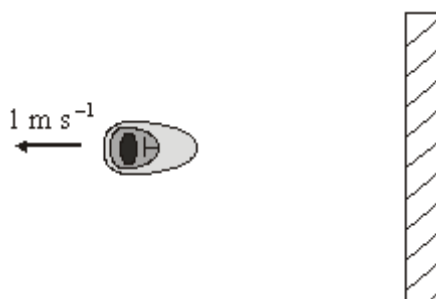
(Total 3 marks)

Q37.

At a fairground, a dodgem car is moving in a straight line, directly towards a wall, with speed 2 m s^{-1} .



It hits the wall and rebounds with a speed of 1 m s^{-1} .



The total mass of the car and passengers is 240 kg and they can be modelled as a single particle.

(a) Find the magnitude of the impulse given to the car as a result of the collision.

(3)

(b) A possible model for the force, F newtons, between the dodgem car and the wall during contact is given by

$$F = 500t(6 - 5t), \quad 0 \leq t \leq 1.2,$$

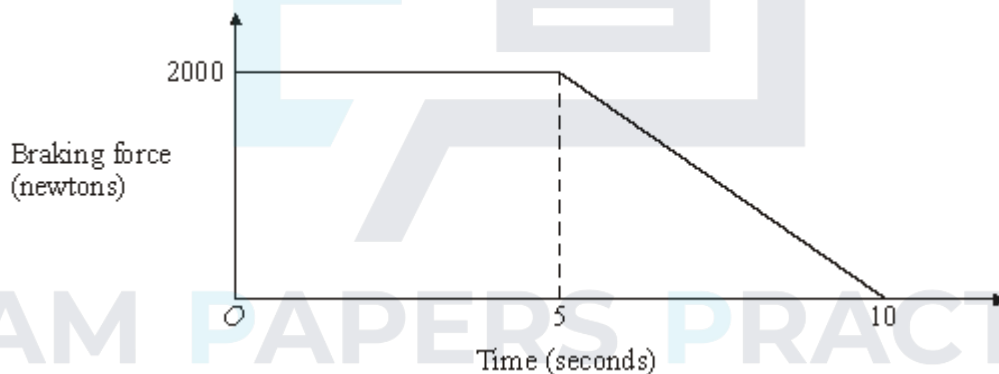
where t denotes time in seconds.

- (i) State the values of t for which the force has zero magnitude, (1)
- (ii) Sketch the graph of F against t . (2)
- (iii) Show that the maximum magnitude of the force using this model is 900 N. (3)
- (iv) Briefly explain how the answer in part (a) relates to the graph in part (b)(ii). (1)

(Total 10 marks)

Q38.

A car is being driven along a straight horizontal road. A braking force is applied to the car to bring it to rest after 10 seconds. It may be assumed that no other resistive forces act on the car during this time. A model for such a braking force is shown by the graph below.

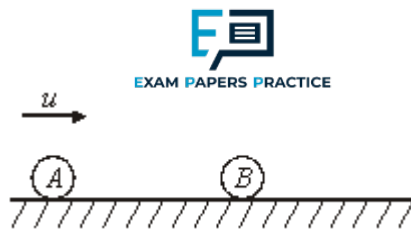


- (a) Show that the impulse acting on the car for the 10 second interval shown is of magnitude 15 000 N s. (3)
- (b) The car has a mass of 1000 kg. Find the speed of the car at the instant when the braking force is first applied. (2)

(Total 5 marks)

Q39.

Two identical smooth spheres, A and B , are each of mass m . Sphere A moves with speed u on a smooth horizontal floor directly towards sphere B , which is at rest. The spheres are shown in the diagram.

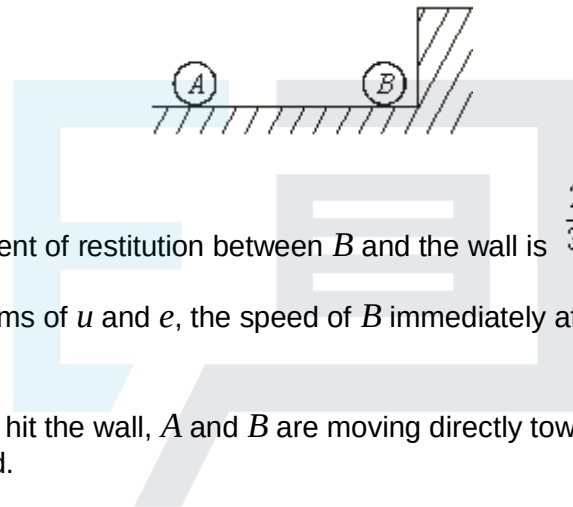


The coefficient of restitution between A and B is e .

- (a) Show that the speed of B immediately after the impact with A is $\frac{1}{2}u(1 + e)$ and find the speed of A , in terms of u and e .

(6)

- (b) Sphere B constantly collides with a vertical wall which is perpendicular to the direction of motion of B .



The coefficient of restitution between B and the wall is $\frac{2}{3}$.

State, in terms of u and e , the speed of B immediately after it hits the wall.

(1)

- (c) After B has hit the wall, A and B are moving directly towards each other with the same speed.

- (i) Show that e , the coefficient of restitution between A and B , is $\frac{1}{5}$.

(2)

- (ii) State the speed of A and B in terms of u .

(1)

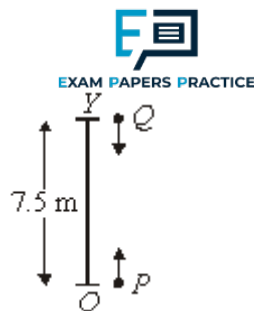
- (d) The first collision between A and B occurs at a distance d from the wall. By modelling the spheres as particles, determine, in terms of d , the distance of the second collision from the wall.

(6)

(Total 16 marks)

Q40.

The point Y is 7.5 metres vertically above the point O . A particle P is projected vertically upwards from O and, at the same time, a particle Q is dropped from Y .



The time, t seconds, is measured from the start of the motion.

When $t = \frac{5}{7}$, P has risen 5 metres.

- (a) (i) Find the distance that Q has fallen when $t = \frac{5}{7}$. (2)

- (ii) Verify that P and Q collide when $t = \frac{5}{7}$. (1)

- (b) The masses of P and Q are 0.2 kg and 0.3 kg respectively. Immediately before the collision, P is moving vertically upwards with speed 3.5 m s^{-1} . The particles coalesce due to the collision. Use the conservation of momentum to find the magnitude and direction of the velocity of the combined particle immediately after the collision.

(6)

(Total 9 marks)

Q41.

Two cars, A and B , are travelling in the same direction on a straight horizontal road. Car A is travelling at 12 m s^{-1} when it collides with car B which is travelling at 8 m s^{-1} , as shown in the diagram.



Car A has mass 800 kg and car B has mass 1000 kg. To model this collision, the cars

can be considered as particles. The coefficient of restitution between these particles is $\frac{1}{8}$.

Show that the speed of car B immediately after the collision is 10 m s^{-1} and find the speed of car A .

(Total 7 marks)

Q42.

Two particles, A and B , of masses $3m \text{ kg}$ and $m \text{ kg}$ respectively, are moving in the same straight line and in the same direction on a horizontal surface. The particles collide

directly. As a result of the collision, the particles coalesce to form a single particle C .

Immediately before the collision, A is moving with speed $v \text{ m s}^{-1}$ and B is moving with speed 2 m s^{-1} . Immediately after the collision, C is moving with speed 1.5 m s^{-1} .

- (a) Find the value of v .

(3)

- (b) The surface on which the particles move is rough. After the collision, C moves with constant retardation and comes to rest at a distance of 3 metres from the point of collision. Find the magnitude of the retardation of C .

(3)

(Total 6 marks)

Q43.

Two particles, A and B , of masses $2m \text{ kg}$ and $m \text{ kg}$ respectively, are moving directly **towards** each other on a smooth horizontal surface. The speeds of A and B are 2 m s^{-1} and 6 m s^{-1} respectively.

The particles A and B collide and subsequently move directly **away** from each other with speeds $3V \text{ m s}^{-1}$ and $V \text{ m s}^{-1}$ respectively.

Find the value of V .

(Total 4 marks)

Q44.

In an accident, a car A collides directly with a stationary car B . The masses of A and B are 1200 kg and 1500 kg respectively. The speed of A immediately before it collides with B is 43.2 km h^{-1} . To model the collision, the cars may be considered as particles. The coefficient of

restitution between these particles is $\frac{1}{4}$.

- (a) Show that the speed of B immediately after the collision is 24 km h^{-1} and find the speed of A .

(7)

- (b) Show that the magnitude of the impulse of A on B due to the collision is $10\,000 \text{ N s}$.

(3)

- (c) During the collision a varying force, F newtons, acts between the cars. In a simple model, F can be assumed to increase uniformly from zero to a maximum value and then decrease uniformly to zero.

- (i) Sketch a graph showing how F varies with time, t seconds.

(2)

- (ii) Given that the cars are in contact for 0.01 seconds, use your graph to find the maximum value of F .

(3)

- (iii) State **one** criticism of this model of the force F .

(1)

(Total 16 marks)

Q45.

A particle P moves with a constant velocity.

- (a) The particle passes through the points with the position vectors $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 8 \end{bmatrix}$ at times $t = 0$ and $t = 3$ respectively. The units of distance and time are metres and seconds respectively.

Show that the velocity of P is $\begin{bmatrix} 1 \\ 2 \end{bmatrix} \text{ m s}^{-1}$.

(3)

- (b) When $t = 3$, the particle P , of mass 0.2 kg collides with a particle Q . The particle Q has mass 0.1 kg and, immediately before the collision, it is moving with velocity $\begin{bmatrix} -5 \\ 5 \end{bmatrix} \text{ m s}^{-1}$.

As a result of the collision, P and Q coalesce into a single particle, R .

Find the velocity of R immediately after the collision.

(4)

- (c) Assuming that the velocity of R remains constant, find the position vector of R three seconds after the collision.

(4)

(Total 11 marks)

Q46.

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A moving particle P collides with a particle Q , which is also moving.

The particle P is of mass 0.25 kg, and its velocities immediately before and after the

collision are $\begin{bmatrix} 3 \\ 6 \end{bmatrix} \text{ m s}^{-1}$ and $\begin{bmatrix} -1 \\ 4 \end{bmatrix} \text{ m s}^{-1}$ respectively.

- (a) (i) Find the momentum **lost** by P due to the collision. State the units of your answer.

(3)

- (ii) State the momentum gained by Q due to the collision.

(1)

- (b) The mass of Q is 0.1 kg.

Immediately after the collision, Q moves with velocity $\left[\frac{5}{2}\right] \text{ m s}^{-1}$.

Find the velocity of Q immediately before the collision.

(3)

(Total 7 marks)

Q47.

Two particles, A and B , are travelling towards each other along a straight line. Both particles have a speed of 5 m s^{-1} . The mass of A is 8 kg and the mass of B is 4 kg .

- (a) If A is brought to rest by the collision, find the speed of B after the collision.

(4)

- (b) If, after the collision, the speed of B is twice the speed of A , and both particles move in the same direction, find the speed of B .

(4)

(Total 8 marks)

Q48.

An arrow of mass m , travelling horizontally with speed $63u$, hits a block of mass $80m$, which is at rest on a smooth horizontal surface. As a result of the impact, the arrow is embedded in the block.

- (a) Find, in terms of u , the speed of the combined body after the impact.

(3)

- (b) Find the number of such arrows which must be fired into the block to make the block move with speed $7u$.

(4)

(Total 7 marks)

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Q49.

Two particles, P and Q , of masses m and $4m$ respectively, lie at rest on a smooth horizontal floor. They are joined by a light inelastic string. Initially, the string is slack.

Particle Q is set in motion with velocity $2u$ away from P .

- (a) Find the speed of Q immediately after the string becomes taut.

(3)

- (b) Find the impulse in the string as it becomes taut.

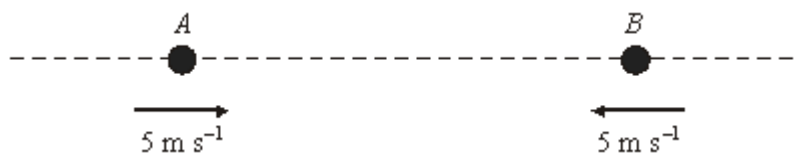
(2)

(Total 5 marks)

Q50.

Two particles, A and B , are moving towards each other along a straight horizontal line. They are both moving at 5 m s^{-1} . The mass of A is 3 kg and the mass of B is 2 kg . The

particles collide.



- (a) If the particles coalesce, find the speed of the combined particle after the collision.
- (b) If the particles do not coalesce during the collision, but the speed of particle A is reduced to 0.5 m s^{-1} and its direction of motion does not change, find the speed of particle B after the collision.

(3)

(4)

(Total 7 marks)

Q51.

A particle P , of mass $6m$, travelling with speed $3u$ collides directly with a particle Q , of mass $2m$, travelling in the opposite direction with speed u .

The coefficient of restitution between the two particles is $\frac{1}{4}$.

- (a) Show that the speed of P after the collision is $\frac{7}{4}u$.

(6)

- (b) Find the speed of Q after the collision.

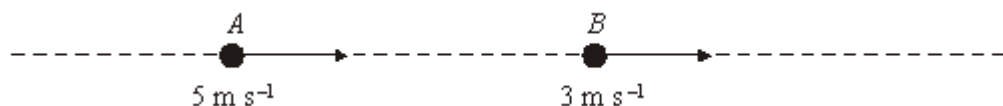
(2)

(Total 8 marks)

Q52.

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Two particles, A and B , are moving with constant speeds in the same direction along a straight horizontal line. The velocity of A is 5 m s^{-1} and its mass is 0.1 kg . The velocity of B is 3 m s^{-1} and its mass is 0.4 kg . The two particles collide. The diagram shows the velocities before the collision.



- (a) If the particles coalesce during the collision, find the velocity of the combined particle after the collision.
- (b) If the particles do **not** coalesce during the collision, and the velocity of B increases to 3.5 m s^{-1} , find the velocity of A after the collision.

(3)

(4)

(Total 7 marks)

Q53.

A smooth sphere, A , has mass $3m$ and velocity $7\mathbf{i} - 8\mathbf{j}$. It collides with a second smooth sphere, B , which has mass m and velocity $2\mathbf{i} + 5\mathbf{j}$. The two spheres have the same radius. After the collision, the velocity of B is $5\mathbf{i} - 4\mathbf{j}$.

- (a) Find the velocity of A after the collision.

(4)

- (b) Find the change in momentum of B .

(2)

(Total 6 marks)

Q54.

Three particles, A , B and C , are initially at rest in a straight line on a smooth horizontal surface. The masses of the particles are 2 kg, 4 kg and m kg respectively.



- (a) Particle A is set in motion with speed 4 m s^{-1} directly towards particle B . After colliding with B , particle A continues to move in the same direction but with speed 1 m s^{-1} . Find the speed of B after this collision.

(3)

- (b) Particle B then collides with particle C . After this collision, C moves with a speed of 2 m s^{-1} . Find the speed of B after this collision, giving your answer in terms of m .

(3)

- (c) If A collides with B again, show that $m > 1$.

(4)

(Total 10 marks)

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Q55.

Three particles, P , Q and R , of masses m , $3m$ and $5m$ respectively, are in a straight line on a smooth horizontal surface. Initially Q and R are at rest and P is travelling with velocity $2u$ directly towards Q . When the distance between P and Q is a , particle Q is struck a blow which results in Q travelling with the same velocity as P .



- (a) Find the impulse of the blow.

(2)

- (b) In the subsequent motion, Q collides directly with R . In this collision, Q is brought to rest.

- (i) Find the velocity of R after the collision.

(3)

(ii) Show that the coefficient of restitution between Q and R is $\frac{3}{5}$. (2)

(iii) Find the time after this collision when P and Q collide. Give your answer in terms of a and u . (2)

(c) Another two particles, A and B , of masses m and M respectively, are on the smooth horizontal surface. Particle A is travelling with velocity u directly towards B , which is at rest. The two particles collide. Particle A is brought to rest by this collision.

Prove that $M \geq m$.

(3)
(Total 12 marks)

Q56.

A child of mass 25 kg is sitting in a trolley of mass 13 kg. The trolley is initially at rest on a horizontal surface. A ball of mass 2 kg is thrown towards the child, who catches it. The ball is travelling horizontally at 5 m s^{-1} just before it is caught. Assume that there is no resistance to the motion of the trolley.

(a) Find the speed of the child and the trolley after the ball has been caught. (2)

(b) The child places the ball in the trolley. A second, identical ball is thrown, in the same direction as the first ball, and the child catches it. This ball is travelling horizontally at 6 m s^{-1} just before it is caught. Find the speed of the child and the trolley after the second ball has been caught. (3)

(Total 5 marks)

Q57.

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Three particles A , B and C are set into motion along a straight line on a smooth horizontal surface.

A has mass 3 kg and initially moves at 3 m s^{-1}

B has mass 2 kg and initially moves in the same direction as A at 2 m s^{-1}

C has mass 5 kg and initially moves in the opposite direction to A and B at 1 m s^{-1}

The initial velocities and positions are shown in the diagram.



(a) The particles A and B collide and join together. Show that the combined particle moves with a speed of 2.6 m s^{-1} after the collision. (2)

(b) When the combined particle collides with the particle C , they join together. Find the

velocity of this new combined particle after this collision.

(3)

- (c) After the second collision a force acts on the new combined particle for 0.2 seconds and brings it to rest. Find the magnitude of this force.

(3)

(Total 8 marks)

Q58.

A particle, A , of mass $4m$ moving with constant speed u collides directly with a stationary particle, B , of mass m . After the collision both particles move in the same direction. The velocity of B is twice that of A .

- (a) Show that the speed of the $4m$ mass after the collision is $\frac{2}{3}u$.

(4)

- (b) Find the coefficient of restitution between the particles in the collision.

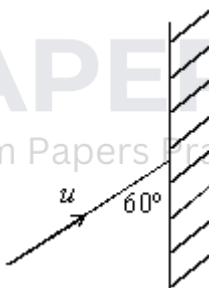
(5)

(Total 9 marks)

Q59.

A sphere of mass m , moving on a smooth horizontal surface, hits a smooth vertical wall. Just before it hits the wall, the sphere is moving at an angle of 60° to the wall with velocity u .

The diagram shows the view from above.



The coefficient of restitution between the wall and the sphere is $3/4$.

- (a) Modelling the sphere as a particle, find the angle through which the direction of motion of the sphere is changed.
- (b) The impulse exerted by the wall on the sphere acts on the sphere for 0.05 seconds. Given that $m = 0.3$ kg, and $u = 5$ m s⁻¹ find the average impulsive force acting on the sphere.

(6)

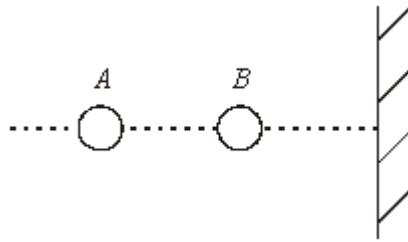
(7)

(Total 13 marks)

Q60.

Two spheres, A of mass $2m$ and B of mass m , are at rest on a smooth horizontal surface.

The centres of the spheres lie on a straight line which is perpendicular to a fixed smooth vertical wall. All motion takes place in the direction of this line.



Sphere A is hit by a blow of impulse J and moves with a speed of $3u$ directly towards the wall. Model the spheres as particles.

- (a) Find J in terms of m and u .

(2)

- (b) Sphere A collides directly with sphere B which is at rest.

The coefficient of restitution in this collision is $\frac{3}{5}$.

Find the speeds of A and B after this impact in terms of u .

(6)

- (c) Sphere B now hits the wall.

The coefficient of restitution between the sphere and the wall is $\frac{1}{2}$.

Find the speed of B after it strikes the wall.

(2)

- (d) Initially sphere B is a distance $4a$ from the wall. Find the distance of B from the wall when B and A collide again.

(7)

(Total 17 marks)

Q61.

Three smooth spheres, A , B and C , of equal radii and masses m , $3m$ and km respectively, lie at rest in a straight line on a smooth horizontal table with B between A and C .

The sphere B is connected to A and C by two light inextensible strings, each of which is initially slack.

The sphere A is then projected directly away from B with speed u .

- (a) Find, in terms of k and u , the common velocity of the three spheres after C has been jerked into motion.

(3)

- (b) Find, in terms of k , m and u , the magnitude of the impulse in each string at the instant when C begins to move.

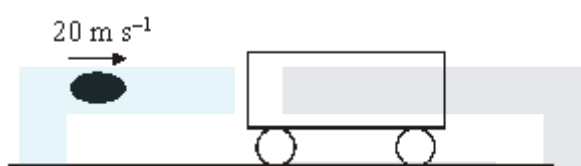
Verify that the ratio of these impulses is independent of k .

(6)

(Total 9 marks)

Q62.

A trolley, of mass 10 kg , is placed at rest on a set of straight horizontal rails. Large pellets, each of mass 0.5 kg , are fired at the trolley. When each pellet hits the trolley, the pellet is travelling horizontally and parallel to the rails at a speed of 20 m s^{-1} . When the pellets hit the trolley, they stick to it and continue to move with the trolley. Assume that there is no resistance to the motion of the trolley.



- (a) Show that the speed of the trolley after it has been hit by the first pellet is $\frac{20}{21}\text{ m s}^{-1}$.

(2)

- (b) Find the speed of the trolley after it has been hit by the second pellet.

(4)

(Total 6 marks)

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